

Standard Form & Indices

Test 9 · Answer sheet · 41 marks total

Section A — Laws of indices

A1. 8^9 (1)

A2. 7^8 (1)

A3. 2^{18} (1)

A4. $2^3 \times 2^3 = 2^6$ (1) = **64** (1)

Section B — Zero and negative indices

B1. **1** (1)

B2. $1 \div 12^2 = 1 \div 144$ (1) = **1/144** (1)

B3. $1 \div 20^1 = 1 \div 20$ (1) = **0.05** (1)

Section C — Writing standard form

C1. 1.8 (1) = **1.8×10^4** (1)

C2. 3.9 (1) = **3.9×10^6** (1)

C3. 5.6 (1) = **5.6×10^{-3}** (1)

C4. 2 (1) = **2×10^{-5}** (1)

Section D — Converting back, and calculating

D1. **94,000** (1)

D2. **0.000003** (1)

D3. $8 \times 5 = 40$, $10^4 \times 10^3 = 10^7$ (1) = 40×10^7 (1) adjust to standard form: **4×10^8** (1)

D4. $5 \div 2 = 2.5$ (1) $10^7 \div 10^3 = 10^4$ (1) already in standard form: **2.5×10^4** (1)

Section E — Powers to know, and estimating roots

E1. **216** (1)

E2. $5^2=25$, $6^2=36$ = between **5 and 6** (1)

Section F — Laws of indices with algebra

F1. **c^{15}** (1)

F2. **k^{13}** (1)

F3. **m^{18}** (1)

F4. $5 \times 6 = 30$ (1) $p^{4+2} = p^6$ (1) = **$30p^6$**

F5. $2^5 = 32$ (1) $q^{3 \times 5} = q^{15}$ (1) = **$32q^{15}$**

Section G — Fractional indices (Higher)

G1. $\sqrt{169} = \mathbf{13}$ (1)

G2. cube root of 729 = **9** (1)

G3. cube root of 8 = 2 (1) $2^4 = \mathbf{16}$ (1)

G4. $\sqrt{100} = 10$ (1) $1 \div 10 = \mathbf{1/10}$ (1)

Marking note: award method marks for a correctly identified front number/power, or correctly applied law of indices, even if the final answer contains an arithmetic error. Accept equivalent fractions/decimals.